MILITARY SPECIFICATION

TEST STAND, HYDRAULIC SYSTEM, GASOLINE ENGINE DRIVEN, TYPE D-5D

This specification is approved for use by the US Army Aviation Systems Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of gasoline engine driven, trailer-mounted, aircraft hydraulic system test stand, designated type D-5D and rated 0 to 10 gpm at 0 to 3,000 psi; or 0 to 5 gpm at 3,000 to 5,000 psi.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

Federal

GG-G-76	Gages, Pressure and Vacuum, Dial Indicating (For Air, Steam, Oil, Water, Ammonia and Chloro-fluro Hydrocarbon and Compressed Gases)
PPP-T-60	Tape, Packaging, Waterproof

MILITARY

MIL-C-104	Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Aviation Systems Command, ATTN: AMSAV-ELS, St. Louis, Missouri 63120-1798, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	Preservation, Method of
MIL-T-704	Treatment and Painting of Materiel
DOD-D-1000	Drawing, Engineering and Associated Lists
MIL-C-1283	Can, Gasoline Military 5 Gallon
MIL-S-1285	Spout, Can, Flexible
MIL-H-5440	Hydraulic Systems, Aircraft, Types I and II, Design and Installation Requirements for
MIL-R-5520	Reservoirs, Aircraft Hydraulic, Non-separated Type
MIL-H-8790	Hose Assembly, Rubber, Hydraulic, High Pressure (3,000 psi)
MIL-H-8795	Hose Assembly, Rubber, Hydraulic Fuel and Oil Resistant
MIL-W-8005	Wheels and Hubs, for Industrial Pneumatic Tires
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent and Resistant
MIL-H-83282	Hydraulic Fluid, Fire Resistant

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirement for the Control of Electromagntic Interference

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MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS25331	Light Assembly, Press to Test Indicator
MS28720	Filter, Aircraft Hydraulic Line Type, 3000 Psi
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS28759	Hose Assembly, Rubber, Hydraulic and Pneumatic (3,000 psi), Flared Tube
MS28777	Packing, O-ring, Hydraulic Back-up
MS28778	Packing, Preformed, Straight Thread Tube Fitting Boss
MS33649	Bosses, Fluid Connection, Internal Straight Thread
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal
MS33657	Fitting End, Standard Dimensions for Bulkhead Flared Tube Connections
MS51336	Lunette & Coupler, Drawbar, Ring

AIR FORCE-NAVY AERONAUTICAL

AN832	Union-flared Tube, 3/8 Bulkhead Universal
AN924	Nut, Tube, Bulkhead & Universal Fitting
AN929	Cap Assembly, Tube, Pressure Seal
AN6236	Filter Element-hydraulic Replaceable Micronic Reservoir Type
AN6249	Valve, 3,000 Psi Hydraulic Check
AN6289	Nut, Tube

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DRAWINGS

AIR FORCE

48B7385	Ring-Cargo Tiedown
48B7796	Ring Assembly-Tiedown, 10,000 LB
49B6564	Ring-Eyebolt, 25,000 LB Cargo Tiedown

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Board of Fire Underwriters

Pamphlet 70 National Electrical Code (Electrical Wiring and Apparatus)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110).

Occupational Safety and Health Act (OSHA) of 1979 Title 29 Code of federal Regulations, Part 1910 and Amendments - Occupational Safety and Health Administration Standards.

(Applications for copies should be addressed to Superintendent of Documents, US Government Printing Office, Washington, DC 20402)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. If first article approval is required, approval and testing shall be in accordance with 4.4. Approval of the first article shall not relieve the contractor of his responsibility to furnish a product in accordance with this specification. Any change or deviation from the approved first article shall be subject to the prior written approval of the contracting officer.

3.2 Components. The test stand shall consist of the following major components:

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Description	See Requirement
a. Hydraulic System	3.9
b. Gasoline engine and accessories	3.10
c. Trailer and housing	3.11

3.3 General. The requirements for materials, fluids, fittings, reservoir venting, hose, tubing, pressure testing, and cleaning of parts and systems specified in MIL-H-5440 as specified for type I apply as requirements of this specification. When the two specifications conflict, this specification shall govern. All electrical wiring shall be permanently marked or identified at each end.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Material

3.5.1 Protective treatment. When materials are used in the construction of the test stand that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic environmental conditions shall be avoided.

3.6 Design. The test stand shall be a self-contained, mobile testing unit providing not less than 10 gpm flow maximum at 3,000 psi diminishing to 5 gpm maximum at 5,000 psi with components arranged as shown on figure 1. The components shall be arranged for maximum ease of operation and maintenance consistent with the versatility and efficiency required to meet the requirements specified herein.

3.6.1 Reliability. The test stand shall have a minimum Mean Time Between Failure (MTBF) of 152 hours with a minimum reliability of 0.967 for a 5-hour mission at a confidence factor of 0.90.

3.6.2 Maintainability. Maintainability requirements shall be in accordance with MIL-STD-470.

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3.7 Construction. The test stand shall be of such construction as to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage and service.

3.7.1 Adjustments and repairs. The test stand shall be so constructed that adjustments, repair, and overhaul are readily accessible to and easily accomplished by personnel of operating units and overhaul bases. The stand shall be so constructed that simple handtools may be used for maintenance and servicing and special tools will not be required. The stand shall be constructed to provide for maximum safety to operating or maintenance personnel.

3.7.2 Transportability. The test stand structures, components, and mounting shall be constructed to withstand, without damage or permanent deformation, the loading specified in MIL-STD-209, class 1 and 2 or Class 3 for Type II or Type III including transport requirements. The test stand shall be externally air transportable by medium lift helicopters utilizing integral lift points and have tiedown capability for rail, air and marine transport.

3.7.3 Fittings. Unless otherwise specified, all fittings and threaded bosses in the hydraulic circuits and components shall conform to MS33649, MS33656, or MS33657. All 5,000 psi pressure AN and MS fittings shall be steel. All bulkhead-type high pressure fittings shall use an AN6289 locknut with a backup gasket conforming to MS28777. All seals used on fittings shall conform to MS28778.

3.7.4 Welding. All welders employed in the fabrication of the test stand shall be certified in accordance with MIL-STD-248.

3.8 Performance. The test stand shall be capable of operating under and shall exhibit satisfactory resistance to the following conditions:

a. Operating temperatures ranging from -20 degrees to +125 degrees F and storage in temperatures ranging form -65 degrees to +160 degrees F.

b. A pressure altitude range from sea level to 6,000 feet.

c. A relative humidity of 95 to 100 percent at an operating temperature of 125 degrees F and a storage temperature of 160 degrees F.

d. Rainfall as encountered in any locale.

e. Exposure to salt fog.

f. Self-induced vibration for all output conditions and vibrations encountered during shipment.

g. Fungus growth as encountered in tropical climates.

h. Sand and dust particles as will be encountered in desert areas.

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3.8.1 Operating positions. The test stand shall function satisfactorily when its normal operating plane is at an angle of 8-1/2 degrees in any direction from horizontal.

3.8.2 Noise level. The noise level of the test stand shall not exceed the values in the frequency ranges specified in 4.6.11.

3.8.3 Turn radius. The test stand shall be capable of executing a full turn to the right and to the left while being towed at a speed of 10 mph and shall not capsize or skid during turns on dry concrete or asphalt. Turning radius shall be 240 inches or less, measured from the center of the rear axle.

3.8.4 Mobility/Towing. The test stand, while loaded with 800 lbs, shall be capable of being towed 50 miles over improved roads at 20 mph and 10 miles over unimproved roads at 10 mph.

3.8.5 Human Factors. The test stand shall be designed so as to incorporate the human engineering criteria specified in MIL-STD-1472.

3.9 Hydraulic system.

3.9.1 High pressure pump. The high pressure pump shall be a variable stroke, variable pressure, axial, piston-type pump with a pressure compensator and handwheel stroke control. The high pressure pump shall be of adequate capacity to meet the requirements specified herein.

3.9.1.1 The pump shall not require more than 20 bhp from the engine when continuously pumping hydraulic fluid conforming to MIL-H-83282 at the rate of 10 gpm and 3,000 psi differential between the low pressure (inlet) gage and the high pressure gage.

3.9.1.2 The volume adjustment shall be capable of limiting flow to any rate from 0 to 10 gpm. The voloume adjustment shall be a limiting device which permits compensator action to maintain pressure at flow rates below any specific volume adjustment setting.

3.9.1.3 The pump shall have an adjustable pressure compensator control capable of regulating the pressure at a constant value with the flow varying from 0 to 10 gpm. The adjustment range of the pressure compensator shall be from 400 to 5,000 psi. A handle shall be provided for manual setting of the pressure compensator, and a locknut to secure the compensator position setting. A pressure compensator shutoff valve shall be provided for turning the pressure compensator off to facilitate adjustment of the relief valve to any value within the pressure range of 400 to 5,000 psi.

3.9.1.4 A relief valve shall be provided on the inlet side of the high pressure pump to return hydraulic fluid to the return system of the test stand. The setting of this relief valve shall be that recommended by the pump manufacturer.

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3.9.1.5 The high pressure pump shall maintain a rated output of 10 gpm at 3,000 psi diminishing to 5 gpm at 5,000 psi at the same time the inlet fluid pressure to the stand is varied throughout the range of 16 inches Hg. absolute to 0 psig pressure. If necessary, a combination of a high pressure pump charged by a low pressure boost pump may be used. If a boost pump is used, the stand shall be capable of full rated output while the inlet pressure to the boost pump is 16 inches Hg. absolute.

3.9.2 Filters. Low and high pressure filters shall be provided and located in the system as shown on figure 1. Each filter or set of filters shall be equipped with a differential pressure switch installed as shown on figure 1 and shall indicate when the pressure drop across the filter reaches a value at which change of the filter element is required. The indication shall be the closing of a switch which illuminates a MS25333-9 red light on the control panel. All filters shall be located to permit easy drainage, air bleed, and replacement of filter elements.

3.9.2.1 High pressure filter assembly. A 5,000 psi filter assembly, shall be provided in the high pressure line as shown in Figure 1. The filter assembly shall meet the envelope requirements of MS28720-12, shall be non by-pass type incorporating a differential pressure switch with actuation at 100 psi differential pressure. The filter element shall be a 3 micron absolute with an efficiency of 95% when tested per MIL-F-81836 and having a 4,500 psid collapse capability. A 10 gpm flow of MIL-H-83282 at 100F will provide a clean pressure drop through the filter assembly of 13 psid maximum. The differential pressure switch shall activate a red indicator light on the instrument panel. The filter housing shall be positioned to permit servicing the filter element without removing or disconnecting any tubing, hose or other component.

3.9.2.2 Low pressure filter. A suction line filter shall be provided and located as shown on figure 1. The filter shall utilize a minimum of one replaceable AN6236-3 or two replaceable AN6236-2 filter elements. Filling and draining plugs and air bleed valves shall be furnished.

3.9.2.3 Case drain filter. A filter shall be provided in the case drain line of the high pressure pump utilizing a replaceable AN6235-2A micronic element.

3.9.3 High pressure relief valve. A high pressure relief valve shall be adjustable by hand from 400 to 5,000 psig. Relief valves shall be located in the system as shown on figure 1. The adjustable high pressure relief valve shall be mounted on the control panel and shall incorporate a locking device so that vibrations cannot change the setting.

3.9.4 Reservoir. A hydraulic fluid reservoir shall be provided and mounted within the stand enclosure. The reservoir shall have adequate capacity to supply at least 20 gallons of fluid to the system when inclined at 8-1/2 degrees from vertical in any direction, plus additional space for a 3-gallon expansion above the normal full level and at least 2-gallon space below the outlet to the system for the collection of moisture and impurities. The bottom

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of the reservoir will be contoured so that all fluid can be drained when the stand is inclined 8-1/2 degrees from vertical in any direction.

3.9.4.1 The reservoir shall conform to the requirements of MIL-R-5520 for obstruction to flow, structural design, shell, seams, structural strength baffles, fluid connections, sump, drain, vent, vent filter, return line inlet, filler unit, finish, and filling time. Adequate clearance around the filler opening shall be provided to allow rapid filling from 1-gallon cans while the stand is in operation.

3.9.4.2 Access plates. One or more access plates, at least 8 x 8 inches, shall be provided to allow visual inspection and manual cleaning of the interior of the reservoir.

3.9.4.3 Drain valve. A drain valve shall be provided for completely draining the reservoir. Necessary clearance shall be provided around the drain outlet to permit rapid draining into 1-gallon containers. The valve shall be 1/2" minimum gate type with soft seat.

3.9.5 Pressure outlet. A pressure outlet shall be provided as shown on Figure 1. The outlet shall be equipped with an Aeroquip Part No TB 155-S4-8D or equal, quick disconnect coupling half and Aeroquip Part NO. 155S7-8D, or equal, dust cap and chain assembly. The end of the chain shall be permanently attached to the frame of the stand. The pressure outlet location shall not allow leakage from the connection to fall on the trailer tires nor within the stand enclosures.

3.9.5.1 Pressure connections. All valves, connections and tubing used from the high pressure pump discharge to the end of the 20 foot pressure hose assembly (with Aeroquip pressure outlet) shall be of a size that will not allow the total pressure drop to exceed 30 psi at 10 gpm flow rate at a fluid temperature of 100 degrees F to 155 degrees F. All connections and fittings on the high pressure side, exclusive of hose assemblies, shall be no less than 3/4 inch tubing size. Tubing shall be used on all high pressure lines within the stand, except the pump outlet to relief line shall be a hose assembly.

3.9.6 Selector valve. A 4-way 2-position open center rotary type selector valve, Teledyne Republic Part No. 744-1-1/2-D2, or equal, shall be provided, shall direct return fluid to the top of the reservoir and allow the pump to draw fluid out of the bottom of the reservoir. In the other position the selector valve shall direct return fluid to the pump inlet and block flow into and out of the reservoir. The selector valve shall not block return flow while it is being reversed.

3.9.7 Return inlet. The return inlet shall be provided with one Aeroquip Part No. 155-S11-12D, or equal, coupling half and dust cap assembly. The free end of the dust cap chain shall be permanently attached to the frame of the stand. The location of the coupling half shall not allow leakage to fall on the trailer tires nor with the stand enclosure.

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3.9.7.1 Low pressure components and connections. The size of all components and connections used in the return side of the hydraulic system shall be not less than 1 inch tubing size. Pressure rating will be of sufficient strength to withstand 100 psi aircraft reservoir pressure when system is in "closed loop" position. "Closed loop" position for this test stand means 4-way, 2-position selector valve (see paragraph 3.9.6) is positioned to bypass the test stand reservoir.

3.9.8 Bypass valve. A 3-way, 2-position bypass valve 3/4 in. tube size or larger shall be provided and mounted on the control panel. The valve will be provided with a handle and shall go from open to full closed in 180 degree rotation or less. The valve shall not require more than 80 inch-pounds to actuate at 5,000 psi.

3.9.9 Pressure check valve. A pressure check valve which is dimensionally interchangeable with an AN6249-12 valve, but suitable for operation at 5,000 psi, shall be located as shown on figure 1 to prevent reverse flow to the high pressure pump.

3.9.10 Flow indicator. A direct-reading means shall be provided to determine actual flow through the high pressure pump within ±1/2 gpm at all conditions of flow and pressure. This may be accomplished by means of an indicator connected to the pump assembly. The flow-measuring device shall indicate actual flow during reduced flow conditions due the compensator action. A flow meter independent of the high pressure pump may be used. If used, the flow adjustment of the pump shall be indicated in 1/2 gpm increments.

3.9.11 High pressure gage. A flush-mounted, high pressure gage shall be provided and shall have a 4" diameter dial or larger and steel-alloy bourdon tube. The gage shall be accurate to within 1 percent, or less, of full scale and shall conform to GG-G-76, except where conflicts exist, the provisions of this specification shall govern. The gage shall provide a range of from 0 to 6,000 psi and shall incorporate a means to restore the accuracy of the gage under all service conditions. The gage shall be the recess type provided with a mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 5/8" beyond the external surface of the panel. All gages shall incorporate a safety blowout device. The safety blowout device shall automatically relieve the gage case pressure should it reach 2 psi due to gage failure. The pointer shall be adjustable without removing the gage from the panel. The gage movement shall be constructed of stainless steel capable of withstanding all conditions of operation, shipments and storage.

3.9.11.1 Mounting. Gages shall be mounted by screws and tapped holes to permit removal by one man from the front of the panel.

3.9.11.2 Connection. Gages shall be connected to the panel with flexible AN- or MS-type hose, or commercial equivalent, of sufficient length to permit removal and installation from the front of the panel.

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3.9.11.3 Gage line. The high pressure gage line shall be provided with a needle shutoff valve, an accumulator-type pulsation dampener, and a gage test fitting. The pulsation dampener shall prevent gage damage from pressure pulsations and pressure shock.

3.9.11.3.1 Shutoff valve. The needle shutoff valve shall be of the panel mounted type with a corrosion-resistant steel stem. The valve shall show the manufacturer's cold-working pressure rating stamped on the valve body with an arrow indicating the flow direction. The valve shall be suitable for working pressure up to 5,000 psi.

3.9.11.3.2 Gage test fittings. The following fittings shall be mounted on the control panel as shown on figure 1, adjacent to the appropriate gage, and the panel shall be marked to indicated that the fittings are gage test connections:

a. An AN832-4D nipple

b. An AN924-4D nut

c. An AN929-4 cap

3.9.12 Low pressure gage. A flush-mounted compound gage shall be corrosion-resistant steel bourbon tubes and a dial of 4 inch diameter or larger. The gage shall be accurate to within 1 percent of full scale range and shall conform to the requirements to GG-G-76 except that where conflicts exist, the provisions of this specification shall govern. The gage shall provide a range of 25 inches Hg vacuum or less to 125 psi pressure or more. The gage shall be red sectored below 20 inches Hg vacuum and above 110 psig. The gages shall be the recess type provided with mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 5/8 inch beyond the external surface of the panel. All gages shall incorporate a safety blowout device. The safety blowout device shall automatically releive the gage case pressure should it reach 2 psi due to gage failure. The pointer shall be constructed or stainless steel capable of withstanding all conditions of operation, shipments and storage. In In all other respects, this gage and mounting shall conform to the requirements specified in 3.9.11.1, 3.9.11.2, and 3.9.11.3.

3.9.12.1 A panel mounted, hand operated selector valve shall be provided to select the gage pressure before and after the low pressure filter, and to shutoff the gage entirely as shown on figure 1.

3.9.12.2 In the event a booster pump is used, the gage specified in 3.9.12 shall be a combination gage with range form 30 inches Hg vacuum to a pressure twice the boost pump relief valve setting, and the selector valve specified in 3.9.12.1 shall provide an additional position to read on the gage the suction pressure to the boost pump.

3.9.13 Temperature gage. A remote reading temperature gage shall be provided and shall indicate the temperature of the fluid entering the high pressure

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pump. The range of the temperature gage shall be -20 degrees to +220 degrees F and shall provide an accuracy of ±5 degrees F, or better. The gage shall have a dial 2-½ inches or larger in diameter and shall be flush mounted. The dial face shall be marked with a red sector at 165 degrees F and above as a warning.

3.9.14 Fluid level indicator. An electrical type fluid level indicator shall be provided to indicate the quantity of fluid in the reservoir. The fluid level indicator shall be accurate within ±1 gallon of actual and shall be clearly visible to the operator standing at the control panel.

3.9.15 Hose assemblies. The following hydraulic hose assemblies shall be provided with each stand

a. One MS28759H1200 Hose Assembly (20 ft length) conforming to MIL-H-8790, except impulse requirements shall be waived.

b. One MS 28741-12-1200 Hose Assembly (20 ft length) conforming to MIL-H-8795, except impulse requirement shall be waived. Hose to be marked at each end with metal tag: "Do not use on high pressure side of system."

3.9.16 Hose accessories. Two Aeroquip Part No. 155-S5-8D coupling halves with union nuts and 155S9-8D dust plugs, or equal, with interchangeable parts, and two Aeroquip Part No. 155-S5-12D coupling halves with union nuts and 155S9-12 dust plugs or equal, with interchangeable parts shall be furnished and installed in each end of its respective hose assembly specified in para 3.9.15. Dust plugs shall be securely attached to the hoses by means of Aeroquip Part No. 51000S10-12, or equal, chain assemblies.

3.9.16.1 Coupling assemblies. One assembly consisting of an Aeroquip Part No. 3205-8 cupling half, or equal, Aeroquip Part No. 3209-8 dust plug or equal, and AN929-A8 dust cap shall be furnished. One assembly consisting of an Aeroquip Part No. 3205-12 coupling half or equal, Aeroquip Part No 3209-12 dust plug or equal, and AN929-A12 dust cap shall be furnished. The stand shall contain storage compartment for storing the coupling assemblies.

3.9.17 Flushing manifold. A flushing manifold to permit flushing of the hose assemblies shall be mounted on the stand. The manifold shall be adjacent to the pressure outlet and return connections. The manifold shall be equipped with an Aeroquip Part No. TB 155-S4-8D, or equal, quick disconnect coupling half, and Aeroquip Part No. TB155-S7-8D, or equal, dust cap and chain assembly one end. The manifold shall be equipped with an Aeroquip Part No. 155-S11-12D, or equal, quick disconnect coupling half and dust cap assembly on the opposite end.

3.9.18 Oil cooler. An oil cooler shall be furnished and installed in the hydraulic circuit as shown on Figure 1. The cooler shall be able to withstand 300 psi. The cooler must be capable of maintaining the oil inlet to the pump temperature below 155 degrees F maximum for a period of 45 minutes with the stand in operation, the compensator set at 5,000 psi, the relief valve at 3,000 psi, the bypass valve fully closed, the pressure inlet and outlet

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capped, and the volume control set at five gpm.

3.10 Gasoline engine and accessories.

3.10.1 Gasoline engine. The gasoline engine shall be a Teledyne Wisconsin Model VG4D or equal.

3.10.2 Gasoline engine accessories. The following items shall be furnished in addition to other items as may be necessary to meet the requirements specified herein. These items shall be of sufficient capacity or quality to meet the performance requirements of this specification.

3.10.2.1 The engine shall be provided with a positive displacement-type fuel pump, fuel filter with glass collecting bowl, oil filter, cooling system, and a carburetor oil bath air intake filter.

3.10.2.2 Exhaust system. An exhaust system shall be provided and shall include an exhaust collector, muffler, and spark arrester. The exhaust system shall meet the code of Federal Regulations, sub part Z, Section 1910.1000 for carbon monoxide.

3.10.2.3 Governor. A single speed governor shall be provided and shall maintain engine speed within five percent of governor setting under all load conditions.

3.10.2.4 Throttle and choke. The engine shall be equipped with a manual throttle to lower the engine speed to idling speed, and a manual choke to assist in starting the engine.

3.10.2.5 An electrical starting and battery charging system of sufficient capacity with regulator control shall be provided. The electrical system shall be 24V (28 ± ½ volts regulated).

3.10.2.6 Battery drain lines. A drain pan and drain line shall be installed beneath the battery to remove spillage and liquid discharge. The drain lines shall be extended far enough to clear the metal frame to the trailer.

3.10.2.7 Circuit breaker. Circuit breakers rated in accordance with Pamphlet 70 shall be installed as required to protect the electrical system and wiring against faults and overload currents.

3.10.2.8 Fuel tank. A fuel tank of adequate capacity to provide fuel for 4 hours or more of full-load operation shall be provided and mounted within the outline of the test stand housing. The fuel tank and supporting frame shall be sufficiently rigid to prevent damage due to vibration and handling. A filler opening of not less than 2 inches inside diameter shall be provided and located to facilitate filling from 5-gallon containers conforming to MIL-C-1283 using a

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flexible spout conforming to MIL-S-1285. The filler opening shall be provided with a screen for removing impurities. The fuel tank cap shall be painted red and stenciled to denote the type of fuel to be used. The fuel tank cap shall be securely attached to the fuel tank by means of a metal chain. A sump shall be provided in the fuel tank suitable for collection and accumulation of not less than 1 pint of water. A drain shall be provided and located at the bottom of the sump to permit complete drainage of the fuel tank. Fittings shall be installed to carry the drainage from the fuel tank to the outside of the stand housing.

3.10.2.9 Fuel gage. A float-type electrical fuel indicator shall be installed in the fuel tank, with the gage placed on the instrument panel underneath the ammeter. A spring-loaded toggle switch shall be placed in the circuit (normal off position) to prevent discharging battery when not in operation.

3.10.2.10 The engine shall be provided with a starter control, tachometer, hourmeter, oil pressure gage, ammeter, and such other instruments and controls as may be required.

3.10.2.11 All engine instruments and controls shall be grouped together on one section of the control panel.

3.10.2.12 Thermoswitch. A theromswitch shall be installed to shutoff the gasoline engine when the fluid into the high pressure pump reaches 165 degrees ±2 degrees F.

3.10.2.13 Oil pressure gage. The engine oil pressure shall be obtained with an oil pressure gage that directly indicates oil pressure. The gage shall be installed on the control panel.

3.11 Trailer and Housing. The components of the test stand shall be mounted and assembled on a 4-wheel trailer and covered with a weather-resistant housing.

3.11.1 The trailer shall be of a 4-wheel, 2-wheel steerable type.

3.11.2 Wheels, tires, and hubs. The trailer shall be mounted on four wheels, hubs, and size 6:90-9 tires in accordance with MIL-W-8005. The tires shall be of 6-ply rating.

3.11.3 The trailer frame shall be of welded steel construction.

3.11.4 Parking brakes. The trailer parking brakes shall hold the test stand in a fixed position during all operating conditions and positions specified herein. Brakes shall be of the internal expanding type capable of locking the rear wheels sufficiently to cause the tires to skid on dry, rough concrete when sufficient force to move the trailer is applied. Brakes for both rear wheels shall be interconnected and shall operate from one handle.

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3.11.4.1 The parking brake control device shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be ¼-inch black letters and shall be visible to the operator.

3.11.5 Springs. Springs shall be furnished on the trailer to provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be so constructed that not more than 2 inches are added to the frame height for spring deflection.

3.11.6 Steering. Ackerman type steering incorporating tie rods and king-pins shall be provided on the trailer to provide a high degree of maneuverability. Grease fittings shall be provided for lubrication of all bearing or metal surfaces having relative movement. Cramping angle shall not be less than 40 degrees.

3.11.7 The trailer shall be furnished with a hinged drawbar assembly suitable for towing the test stand 20 mph behind a standard military 2-½ ton 6 by 6 truck, Type M35. The drawbar shall be equipped with a lightweight Lunnette eye conforming to MS51336 with a 3 inch inside diameter and a 1-5/8 inch diameter cross section.

3.11.8 The trailer shall support 800 pounds in addition to the test stand components, instrument panel, and housing.

3.11.9 The trailer shall be provided with four or more lifting and tiedown fittings conforming to Drawings 48B7385, 48B7796, or 49B6564, capable of withstanding the loadings specified in 3.7.2.

3.11.10 The fully-loaded trailer shall provide at least 8 inches of road clearance and tread width shall be 49 inches.

3.11.11 Control panel. A control panel shall be provided for mounting all the meters, valves, indicators, gages, switches, and controls furnished with the test stand. Unless otherwise specified, all meters, valves, indicators, gates switches, and controls shall be mounted in the control panel The control panel shall be constructed of No. 12 US gage (0.1046 inch) sheet steel, or thicker, and supported by a welded steel self-supporting framework. The control panel frame shall be bolted or welded to the trailer bed and so positioned that the trailer wheels, tow bars, et cetera, will not interfere with the operator. The pump pressure control and volume control may be mounted on the pump if they are located in such positions that they may be actuated by the operator while facing the control panel. The control panel shall be located at a sufficient height above the ground and inclined at a sufficient angle, 10 degrees minimum, to facilitate convenient observation and operation of all items by an operator 5 feet 10 inches tall. Instruments subject to error due to vibration shall be shock mounted. All items mounted on the control panel shall be identified as

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to use and cautions by etched or engraved metal or plastic plates permanently attached to the panel.

3.11.11.1 A metal or plastic, stamped or engraved, permanent-type caution sign displaying the following warning in ¼-inch letters, or larger, shall be provided and securely attached in the center of the control panel:

(CAPACITY. 10 GPM FLOW MAXIMUM AT 3,000 PSI DIMINISHING TO 5 GPM FLOW MAXIMUM AT 5,000 PSI. ALWAYS SET THE HIGH PRESSURE RELIEF VALVE AT A HIGHER PRESSURE (50-PSI HIGHER) THAN THE PUMP COMPENSATOR PRESSURE - DO NOT USE THE RELIEF VALVE TO REGULATE TEST STAND OUTPUT PRESSURE).

3.11.11.2 Diagram plate. A 4- by 6-inch, engraved, permanent-type diagram plate, arranged similarly to figure 1 to show the hydraulic circuit of the stand, shall be securely fastened the instrument panel.

3.11.12 Housing. The test stand components and control panel shall be completely enclosed and protected by a weather-resistant housing. The housing shall consist of a welded-steel framework covered with No. 16 US gage (0.0598 inch) sheet steel, or heavier, with doors and adjustable louvers as required. The housing shall support a 200-pound weight evenly distributed over an area 4.5 by 5 inches and applied in any location on the top of the stand. The housing shall be braced to prevent distortion or excessive vibration during handling and operation of the stand or handling of the housing while removed from the stand. The stand shall be so designed that water or hydraulic fluid spilled in any place within the housing will drain to the exterior and will not collect within the enclosure. Metal hooks shall be furnished on the exterior of the housing to permit temporary storage of the pressure hoses. The hooks shall be designed to permit flat positioning against the housing when not in use.

3.11.12.1 Internal accessibility and ventilation. The housing shall be provided with access doors and ventilating doors or adjustable louvers to permit access for operating, servicing, and ventilating the stand. All doors or louvers in the housing shall be free from binding and shall be easy to operate with one hand. Provisions shall be incorporated to secure the doors rigidly in the full-open or full-closed position.

3.11.12.2 The housing shall be easily removable by actuating not more than 10 quick-acting disconnecting fasteners and lifting or sliding the entire housing off the remainder to the stand assembly. Handles shall be provided for this purpose. It shall not be necessary to remove the housing to service or ventilate the test stand.

3.11.12.3 The housing shall meet the human factors requirements of MIL-STD-1472, paragraph 5.13.4.6 of 140°F maximum for inadvert contact and 120°F maximum during handling.

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3.12 Compartments. A handbook compartment, 2 by 9 by 12 inches, or larger, shall be provided and permanently and securely attached within the housing. Provisions shall be made for storage of the hose assemblies specified in 3.9.15 within the cutline of the housing. The compartment may be constructed on the trailer bed or may be fastened to the housing assembly.

3.13 Part number of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.14 Electromagnetic interference. The test stand shall meet the requirements of MIL-STD-461 for Class III equipment.

3.15 Dimensions. The length and width of the test stand and housing shall not at any point extend beyond the outline of the wheels. The overall height of the test stand shall not exceed 58 inches with the tires at maximum rated pressure and the fuel tank and hydraulic reservoir drained.

3.16 Weight. The weight of the test stand, including the trailer assembly, shall not exceed 2,500 pounds.

3.17 Finishes and protective coatings.

3.17.1 Priming and painting. Priming shall be accomplished in accordance with MIL-T-704. The top coat shall be green 383 in accordance with MIL-C-46168.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following special characteristics marking shall appear on the nameplate:

RATING 0 TO 10 GPM AT 0 TO 3,000 PSI OR 0 TO 5 GPM AT 3,000 to 5,000 PSI
TOWING SPEED - 20 MPH MAX

3.18.1 Instruction Plates. The test stand shall be provided with instruction plates permanently mounted in the appropriate locations. Instruction plates shall include all essential operational and safety procedures, cautions and acceptable fluids.

3.19 Worksmanship.

3.19.1 General. The test stand, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws, bolts, et cetera. All

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hydraulic hoses shall be routed so as to prevent chaffing with stand offs being used where appropriate.

3.19.2 Riveting. Riveting operations hsall be carefully performed in insure that the rivets are tight and satisfactorily headed.

3.19.3 Cleaning. The test stand shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign materials shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspectons set forth in the specification where such inspections are deemed necessary to assure supplies and services convorm to prescribed requirements.

4.2 Classification of test. The inspection and testing of the test stand shall be classified as follows

a. First article.....See 4.4

b. Quality conformance.....See 4.5

4.3 Test Conditions.

4.3.1 Standard atmosphere conditions. Unless otherwise specified, the test shall be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 73 degrees F).

4.3.2 Test periods. Unless otherwise specified, any test run shorter than one hour shall not be credited as test time.

4.3.3 Fluid flow. The calibration and performance checks of the displaced fluid shall be determined by means of a calibrated flowmeter with certified accuracy of one percent of reading or better.

4.3.4 Preliminary run-in and calibration. Any preliminary run-in of the test stand as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the stand for acceptance or preproduction etc., shall be submitted to the procuring activity for entry in the log book. tests. A certified log of all preliminary running time, calibration data, etc., shall be submitted to the procuring activity for entry in the log book.

4.4 First Article Test.

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4.4.1 First article test. The first article tests shall consist of all tests described under 4.6. The contractor shall submit a test plan for approval by the contracting officer, prior to fabrication of the First Article.

4.4.2 Test report. After the preproduction tests are completed, three complete copies of a test report in accordance with MIL-STD-831 shall be supplied to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the first article test report (same quantity and type of copies) or shall be included as part of that report.

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Description of the engineering reasoning of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Description of the engineering reasoning verifying corrections made, to be made on production items, or proposed to be made that substantiate the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. Estimates (±5 percent of actual experience) of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to permit comparison to the personnel anticipated in similar field work.

4.5 Quality conformance. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling plan and tests.

4.5.1 Individual tests. The test stand shall be subjected to the following tests as described under 4.6. These tests shall be performed on all units produced.

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a. Examination of product.

b. Operational checks

4.5.2 Sampling plan and tests.

4.5.2.1 Lot. A lot shall consist of stands manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.5.2.2 Sampling plan. The following number of sample test stands shall be selected at ramdom from each log and subjected to the semi-endurance test as described under 4.6:

No. of Stands in Lot	Quantity to be Treated
50 or less	1
51 to 150	2
Over 150	3

4.5.2.2.1 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be with-held until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After corrections, all of the sampling tests shall be repeated.

4.5.2.2.2 Individual test may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot or lots produced shall not be made until it is determined that the lot meets all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test Methods.

4.6.1 Examination of product. The test stand shall be inspected to determine compliance with requirements specified herein with respect to materials, workmanhip, and markings.

4.6.2 Operational check. The constant flow operation of the test stand shall be checked at 800, 1,500, 2,500, and 3,000 psi at each rate of low of 2, 4, 6, and 10 gpm and 5,000 psi at 5 gmp with compensation turned off. This test shall be conducted first with the reservoir in the system, and then repeated

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with the reservoir bypassed and with the suction restricted to create a suction of 16 inches Hg absolute pressure at the inlet to the high pressure pump (or 16 inches Hg absolute to the booster pump). The pressure gage shall be accurate to within ±1 percent of full scale at all settings. The graduated flow control handwheel setting shall be accurate within ±½ gallon of actual flow, the flowmeter shall indicate flow within ±½ gallon of actual flow, and the reservoir fluid level gage shall be checked for accuracy of ±1 gallon throughout its range.

4.6.2.1 The operation of the test stand shall be checked at 5 gpm and at 10 gpm with compensation set a 1,500 psi and 3,000 psi respectively. The hydraulic pump shall put out full 5 and 10 gpm flow at all loading pressures below the 1,500 and 3,000 psi respectively, but flow shall fall to nearly zero as soon as the loading pressure reaches compensation pressure setting.

4.6.2.2 The temperature of the fluid entering the high pressure pump on the test stand shall be raised above 155 degrees F and the thermoswitch shall shut off the gasoline engine before the fluid temperature reaches 165 degrees F (±2 degrees F), but not before it reaches 155 degrees F.

4.6.3 Semi-endurance. The test stand shall be subjected to 20 percent of the endurance test time specified in 4.4.7. At the completion of this test, the stand shall then satisfactorily pass the test specified in 4.6.2 and 4.6.2.1.

4.6.4 Oil Cooler. Operate the test stand for 45 minutes under conditions as follows:

Compensator valve set at 5000 PSI.
Relief valve set at 2000 PSI.
Bypass valve fully closed.
Pressure inlet capped
Pressure outlet capped
Volume flow set at 5 gal per minute.

The oil cooler shall be capable of maintaining the oil inlet temperature to the pump below 155°F maximum.

4.6.5 Initial disassembly inspection. The test stand submitted for preproduction tests shall be disassembled to the maximum extent practical and completely inspected, the breakdown weights determined, and measurements made which may used to determine wear. All gages, meters, and instruments shall be laboratory checked against certified calibrating equipment and failure of any item to meet the accuracy requirements shall be cause for rejection. All requirements which may be determined by weight or dimensional measurements and inspection shall be checked for compliance. Failure to meet any requirement shall be cause for rejection. Following this inspection, the stand shall be reassembled and subjected to the remainder of the preproduction tests.

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4.6.6 Environmental test. The test stand shall be subjected to the following environmental tests in accordance with the specified procedures of MIL-STD-810. Unless otherwise specified, upon completion of each test, the stand shall be subjected to the test specified in 4.6.2 and 4.6.2.1. Failure of the stand to operate as required during or after each exposure, failure of any item or assembly due to deterioration from exposures in any test, or formation of any undesirable condition which cannot be remedied by routine maintenance procedures shall be cause for rejection.

4.6.6.1 Low temperature. The test stand shall be subjected to low temperature in accordance with method 502, procedure I or MIL-STD-810, except the storage temperature shall be -65 degrees F and operating temperature -20 degrees F. The test circuit shall include one MS28759H1200 pressure hose, a restrictor valve to build up 3,000 psi pressure, one reservoir filled with not less than 40 gallons of hydraulic fluid conforming to MIL-H-83282 and one MS287741-12-1200 return line. The entire test circuit shall be contained within the cold chamber and stabilized to -20 degrees F prior to start of test. The temperature of the fuel within the test stand fuel tank shall be measured and recorded. The temperature shall be -20 degrees F±3 degrees prior to start of test. Prior to starting or during the test, the fluid in the reservoir shall not be heated from any external source, The engine shall start within 30 minutes or less of the first attempt to start. After starting, the flow and pressure shall be slowly increased to the rated output of 10 gpm and 3,00 psi. The stand shall attain rated output within 30 minutes after starting and shall be operated at rated output for not less than 13 minutes. All controls, gages, meters, valves, systems, and accessories shall be checked for satisfactory operation while operating in the ambient temperature.

4.6.6.2 High temperature. The test stand shall be subjected to high temperature in accordance with method 501, procedure I of MIL-STD-810. Upon completion of the exposure to high temperature, the ambient temperature shall be lowered to 125 degrees F. The constant flow operation of the stand shall be checked at 800, 1,500, and 3,000 psi at each rate of 3, 6, and 10 gpm. The pressure gage shall be accurate to within ±1 percent of full scale at all settings. The graduated flow control handwheel setting shall be accurate to within ±½ gpm of actual flow. After this test, the stand shall be inspected for leakage of fluid.

4.6.6.3 Salt fog. The test stand shall be subjected to salt fog in accordance with method 509, procedure I of MIL-STD-810.

4.6.6.4 Dust. The test stand shall be subjected to sand and dust in accordance with method 510, procedure I of MIL-STD-810.

4.6.6.5 Humidity. The test stand shall be subjected to humidity in accordance with method 507, procedure I of MIL-STD-810.

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4.6.6.6 Rain. The test stand shall be subjected to rain in accordance with method 506, procedure I of MIL-STD-810. The stand shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for 1 hour and while still in the rain test, the stand shall be started and operated at 10 gpm and 3,000 psi for 30 minutes. Failure of the stand to start or operate satisfactorily during the rain test shall be cause for rejection. Upon completion of the rain test, all surfaces which might collect moisture shall be inspected to determine adequacy of drainage facilities and any collection of water in excess of 1/8 inch deep or which could be detrimental to the performance of the stand shall be cause for rejection.

4.6.6.7 Fungus. The test stand shall be subjected to fungus in accordance with method 508, procedure I of MIL-STD-810.

4.6.7 Endurance. The test stand shall be connected through the hoses specified in 3.9.15 to a test rig which shall include a reservoir, restrictors to create test pressures in each high pressure line, a restrictor to create a vacuum in the return line, and a cooler to maintain fluid temperature at approximately 130 degrees F. It shall be subjected to a 300 hour endurance test as specified herein. The ambient air temperature shall be maintained at 75 degrees F or higher during the test. Readings shall be recorded every hour of all gages, meters, and instruments of the stand. Output fluid flow, temperature and pressure, and inlet fluid temperature and pressure shall be measured as specified in 4.3 and recorded. The test procedure shall be as follows:

a. Conduct the tests specified in 4.6.2.

b. fifty hours of operation at 10 gpm and 3,000 psi with the reservoir bypassed and return line shall be pressured to 100 psig.

c. Fifty hours of operation at 5 gmp and 5,000 psi and 100 psig inlet pressure on the test stand return line. Oil flow shall be through the reservoir for 25 hours and bypass the reservoir for 25 hours.

d. Fifty hours of operation at 10 gpm and 3,000 psi with the inlet pressure to the high pressure pump (after low pressure filter) reduced to 16 inches Hg absolute pressure, reservoir bypassed. If a boost pump is used, the inlet pressure to the boost pump shall be held at 16 inches Hg absolute pressure.

e. Fifty hours of operation at 10 gpm and 1,500 psi, reservoir bypassed.

f. Fifty hours of operation at 2 gpm and 3,000 psi, oil flow through reservoir.

g. Fifty hours of operation at 5 gpm and 800 psi, oil flow through reservoir.

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h. Conduct the test specified in 4.6.2. Comparison of these data recorded as specified above shall indicate that no failure or undue wear has resulted from the endurance test.

i. Operate the test stand at 10 gpm and 3,000 psi high pressure pump inlet pressure for 16 inches Hg absolute to 75 psig. Failure of the stand to deliver rated flow or loss of hydraulic fluid from the system shall be cause for rejection.

4.6.8 Thermoswitch. Operation of the thermoswitch shall be checked at least twice. The thermoswitch shall shutoff the gasoline engine when the fluid temperature reaches 165 degrees ±2F.

4.6.9 Altitude. The test stand shall be continuously operated for 5 hours at a pressure altitude of 6,000 feet, or higher, and shall deliver 10 gmp at 3,000 psi during the test. The inlet pressure shall be 16 inches Hg absolute at the inlet to the boost pump, if a boost pump is used. The gasoline engine shall show not signs of overload or overheating during the tests.

4.6.10 Electromagnetic interference. The test stand shall be tested as a complete unit for radiated interference in accordance with MIL-STD-461, class III equipment.

4.6.11 Noise level. The noise level shall be measured 10 feet from the perimeter of the test specimen and shall not exceed the noise level shown below. Measurements shall be taken out-of-doors on a paved surface from at least eight different directions, 45 degrees apart, in a horizontal plane parallel to the trailer bed and passing through the centerline of the high pressure pump drive shaft. During this test, the test stand shall be operating at full rated load.

Cycles per Second	Not Exceeding
20 to 75	108 db
75 to 150	90 db
150 to 300	90 db
300 to 600	83 db
600 to 1,200	78 db
1,200 to 2,400	75 db
2,400 to 4,800	72 db
4,800 to 10,000	71 db

4.6.12 Operation in the tilted position. The test stand shall rest on a platform and shall be kept from rolling by the parking brakes only. The platform supporting the stand shall be tilted in eight different directions 45 degrees apart, on a 8½ degree incline with the true horizontal plane. The stand shall be started and operated at 10 gpm and 3,000 psi for 15 minutes in

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each position. The stand shall perform satisfactorily in every respect and the parking brakes shall prevent the stand from moving on the platforms. While tilted in each of the eight positions, 20 gallons of fluid shall be pumped from the reservoir and then replaced to check capacity specified in 3.9.4.

4.6.13 A 200-pound weight shall be applied slowly and uniformly to an area 4½ by 5 inches in at least six random positions on top of the stand housing, including a reading in the center of each panel. The stand housing shall not be permanently deformed in any way by the application of the weight.

4.6.14 The test stand shall be loaded with 800 pounds of weight so distributed that 400 pounds rests uniformly over the top of the housing and 400 pounds are supported by the trailer frame. The stand shall be towed a distance of 50 miles over improved roads at a speed of 20 mph and a distance of 10 miles at 10 mph over unimproved roads. The 800 pounds of weight shall be removed and the stand shall execute 10 full turns to the right and 10 full turns to the left at an average speed of 10 mph. The stand shall not capsize or skid during turns on dry concrete or macadam. The turning radius for these turns shall be 240 inches measured from the center or the rear axle. Upon completion of the road tests, the stand shall be subjected to the test specified in 4.6.1 and 4.6.2.

4.6.15 Serviceability. The test stand shall be inspected and evaluated from the standpoint of each maintenance, servicing, and operation. Particular note shall be made with regard to provisions to prevent accumulation or dirt, snow, ice et cetera, which may hinder servicing and operation.

4.6.16 Final disassembly inspection. The test stand shall be disassembled to the maximum practical extent for a thorough examination of all parts to disclose failure, distortion, excessive wear, or other unsatisfactory conditions. If measurements are made to determine wear, they shall be compared with similar measurements made during initial disassembly inspection. Stands not conforming to the requirements specified here shall be rejected.

4.6.17 Tiedown and lifting tests. The test stand shall be subjected to the loadings specified in MIL-STD-209. The stand shall be secured during this test by means of the lifting and tiedown fittings only. Upon completion of this test, the stand shall be inspected to determine adequacy of all lifting and tiedown fittings.

4.6.18 Carbon Monoxide. With the test stand engine operating at 1900 RPM in an open (no winds) area, the test stand shall be tested for carbon monoxide emissions at the operator position in accordance with the code of Federal Regulations, subpart Z, section 1910.1000 for carbon monoxide.

4.6.19 Housing/Controls Temperature. The test stand shall be operated for 45 minutes at 3000 PSI, hydraulic fluid flow of 10 GPM and the reservoir bypassed.

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At the end of the 45 minutes operating period take sufficient temperature readings of the test stand housing and operating controls to demonstrate compliance with the temperature requirements of paragraph 3.11.12.3.

4.6.20 Reliability demonstration and test. Satisfactory completion of all test specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.21 Maintainability verification. Verification of the maintainability requirements shall be in accordance with MIL-STD-470.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine compliance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Test stands shall be packaged on each Method IIa of MIL-P-116, and as follows:

5.1.1.1 Hose shall be coiled to the minimum safe diameter and securely bound in three places with cotton tape. The bound coils shall be securely fastened to the stand within the moisture impervious barrier.

5.1.1.2 The drawbar may be detached from the stand and strapped to the stand.

5.1.1.3 The hydraulic system shall be completely drained and all gages turned to the "off" position for delivery. All vent openings of the hydraulic system shall be plugged and sealed with tape conforming to PPP-T-60 to prevent contamination of the system.

5.1.1.4 The tires shall be inflated to 10 pounds above the recommended pressure.

5.1.1.5 The entire fuel system of the engine, including the carburetor, fuel pump, strainer, and fuel lines shall be drained.

5.1.1.6 The battery cables shall be detached from the battery. The cables shall be secured to the battery carrier or other parts with tape conforming to PPP-T-60 in such a manner to prevent scuffing, rubbing, or contact with the battery.

5.1.1.7 The storage battery shipped with the test stand shall be new and dry, charged or uncharged. Plugs shall be left in the filler cap vent holes. A

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waterproof tag shall be secured to the battery indicating the dry condition and procedures required to put the battery in operation condition.

5.1.2 Level C. The test stand shall be packaged in a manner to prevent deterioration or damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A or C, as specified (See 6.2).

5.2.1 Level A. The test stand packaged per 5.1, shall be packed in a crate conforming to MIL-C-104.

5.2.2 Level C. The test stand packaged per 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery at the lowest applicable rate.

5.3 Marking. In addition to the requirements of the contractor order, shipments shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Test Stand, Hydraulic System, Gasoline Engine Drive, Type D-5D

6. NOTES

6.1 Intended use. Type D-5D test stand is intended for use in ground checking and maintenance of aircraft hydraulic systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether the first article approval is required (see 3.1).

c. Level of preservation and packaging, and packing required (see 5.1 and 5.2).

d. Whether special marking is required (see 5.3).

6.3 Definitions. For purposes of this specification, the following definitions will apply:

6.3.1 Improved roads. An improved road is a road of level concrete, macadam, steel landing mats, or smooth, hard-packed dirt.

6.3.2 Unimproved roads. An unimproved road is a road consisting of level-to-rolling gravel trails.

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6.3.3 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of non-failure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expresses as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

6.3.4 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failure.

6.3.5 Disassembled to the maximum practical extent. Disassembled to the maximum practical extent is defined as disassembly of all moving contacting surfaces which have been assembled by the manufacturer of the stand and which are subject to wear (gears, bearings, control mechanisms, et cetera).

Custodian:

Army - AV

Preparing activity:

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(Project No. 4920-A389)

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HYDRAULIC DIAGRAM SCHEMATIC SHT

1. PRESSURE INLET
2. CHECK VALVE
3. FLOW METER
4. SELECTOR VALVE
5. RESERVOIR
6. DRAIN VALVE
7. FILL & VENT
8. BOOST PUMP (IF REQUIRED)
9. HEAT EXCHANGER
10. LOW PRESSURE FILTER
11. LOW PRESSURE GAGE
12. PUMP INLET RELIEF VALVE
13. THERMO SWITCH
14. TEMPERATURE GAGE
15. MAIN PUMP
16. PRESSURE ADJUSTMENT
17. VOLUME ADJUSTMENT
18. NEEDLE VALVES
19. CHECK VALVES
20. CASE DRAW FILTER
21. PULSATION DAMPER
22. GAGE TEST FITTING
23. HIGH PRESSURE GAGE

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HYDRAULIC DIAGRAM SCHEMATIC SHT 2 (CONT)

24. HIGH PRESSURE RELIEF VALVE

25. HIGH PRESSURE FILTER

26. DIFFERENTIAL PRESSURE INDICATOR

27. BYPASS VALVE

28. PRESSURE OUTLET

29. MANIFOLD CONNECTIONS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-7612D	Test Stand, Hydraulic System, Gasoline Engine Driven Type D-5D

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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